Monomial f(q) = a*q^3 Report

Polynomial

$$f(q) = aq^3$$

Naive

$$g_{\text{naive}}(X) = ax^3$$

$$\mathbb{E}[g_{\text{naive}}(X)] = aq^3 + 3aq\sigma^2$$

$$\text{Var}(g_{\text{naive}}(X)) = 9a^2q^4\sigma^2 + 36a^2q^2\sigma^4 + 15a^2\sigma^6$$

$$\text{MSE}(g_{\text{naive}}) = 9a^2q^4\sigma^2 + 45a^2q^2\sigma^4 + 15a^2\sigma^6$$

$$\text{Bias}(g_{\text{naive}}) = 3aq\sigma^2$$

Unbiased

$$g_{\text{unb}}(X) = -3a\sigma^2x + ax^3$$

$$\mathbb{E}[g_{\text{unb}}(X)] = aq^3$$

$$\text{Var}(g_{\text{unb}}(X)) = 9a^2q^4\sigma^2 + 18a^2q^2\sigma^4 + 6a^2\sigma^6$$

$$\text{MSE}(g_{\text{unb}}) = 9a^2q^4\sigma^2 + 18a^2q^2\sigma^4 + 6a^2\sigma^6$$

Comparison

$$\text{mean gap} = -3aq\sigma^2$$

$$\text{variance gap} = -18a^2q^2\sigma^4 - 9a^2\sigma^6$$

$$\text{MSE gap} = -27a^2q^2\sigma^4 - 9a^2\sigma^6$$

$$\text{Bias}(g_{\text{naive}})^2 = 9a^2q^2\sigma^4$$